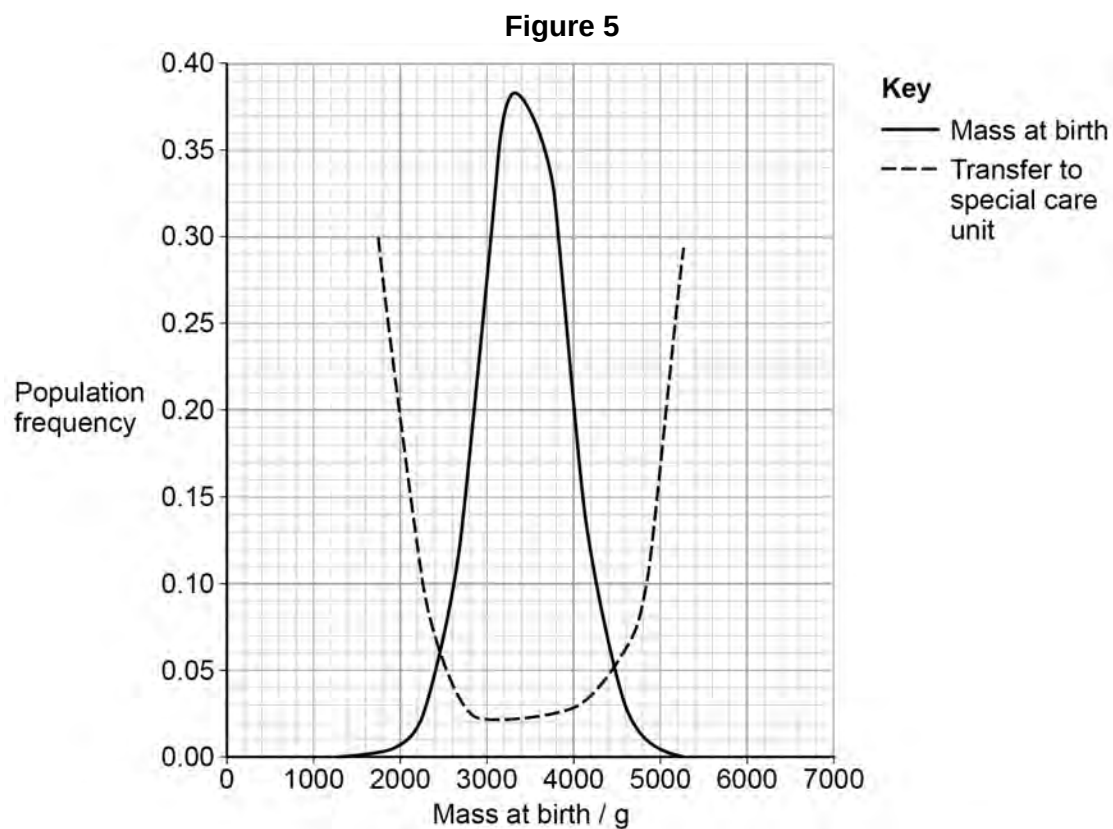


Scientists collected data on 800 000 human births. The data showed the mass of each baby at birth and whether the baby needed to be transferred to a special care unit for very ill babies.

Their results are shown in **Figure 5**.



[3 marks]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Turn over ►

0 4 . 2

The scientists studied the effect of one form, *KIR2DS1*, of the human *KIR* gene on mass at birth.

In the following passage the numbered spaces can be filled with biological terms.

KIR2DS1 is an (1) of the *KIR* gene, found at a (2) on chromosome 19. *KIR2DS1* is 14 021 bases long and is (3) into mRNA that is 1101 bases long. This mRNA is then (4) into a polypeptide 304 amino acids long. The polypeptide is then modified in the organelle, (5), before forming its functional (6) protein structure.

Write the correct biological term beside each number below, that matches the space in the passage.

[3 marks]

- (1) _____
- (2) _____
- (3) _____
- (4) _____
- (5) _____
- (6) _____

0 4 . 3

The scientists studied 1500 more births. They recorded the mass at birth of each baby and the nature of the *KIR* gene in the mother's genome.

Some of their results are shown in **Table 2**.

Table 2

Presence or absence of <i>KIR2DS1</i> in mother's genome	Number of babies with mass at birth:	
	between 2500 g and 4500 g	above 4500 g
Present	389	148
Absent	606	173

The scientists used a statistical test to test the following null hypothesis:

'The presence of *KIR2DS1* in the mother's genome does **not** affect the frequency of births above 4500 g'

Tick (✓) **one** box that gives the name of the statistical test that the scientists should use with the data in **Table 2** to test this null hypothesis.

[1 mark]

Chi-squared

☐

Correlation
coefficient

☐

Student's t-test

☐

Question 4 continues on the next page

[3 marks]
